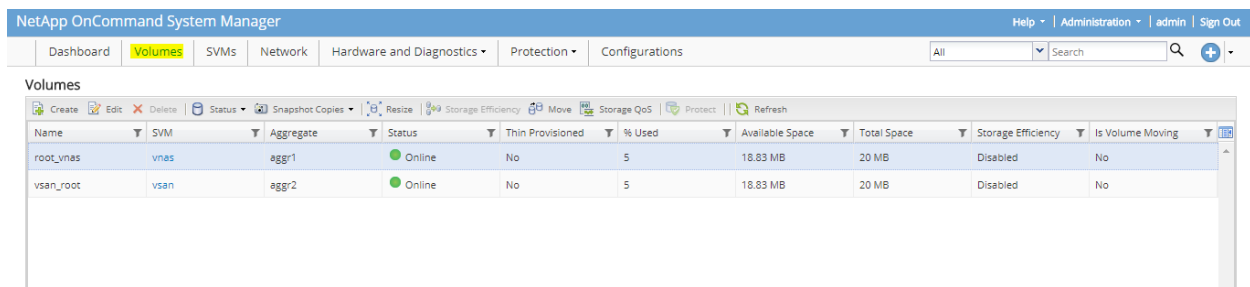


Volume Management

Listing the volumes.

```
P90-C::> volume show
Vserver    Volume      Aggregate    State    Type    Size    Available    Used%
-----
P90-C-01   vol0        aggr0        online   RW      1.79GB   1.14GB      36%
vnas       root_vnas   aggr1        online   RW      20MB     18.83MB     5%
vsan       vsan_root   aggr2        online   RW      20MB     18.83MB     5%
3 entries were displayed.
```

Listing the volumes using System Manager.



Name	SVM	Aggregate	Status	Thin Provisioned	% Used	Available Space	Total Space	Storage Efficiency	Is Volume Moving
root_vnas	vnas	aggr1	Online	No	5	18.83 MB	20 MB	Disabled	No
vsan_root	vsan	aggr2	Online	No	5	18.83 MB	20 MB	Disabled	No

Creating a new volume.

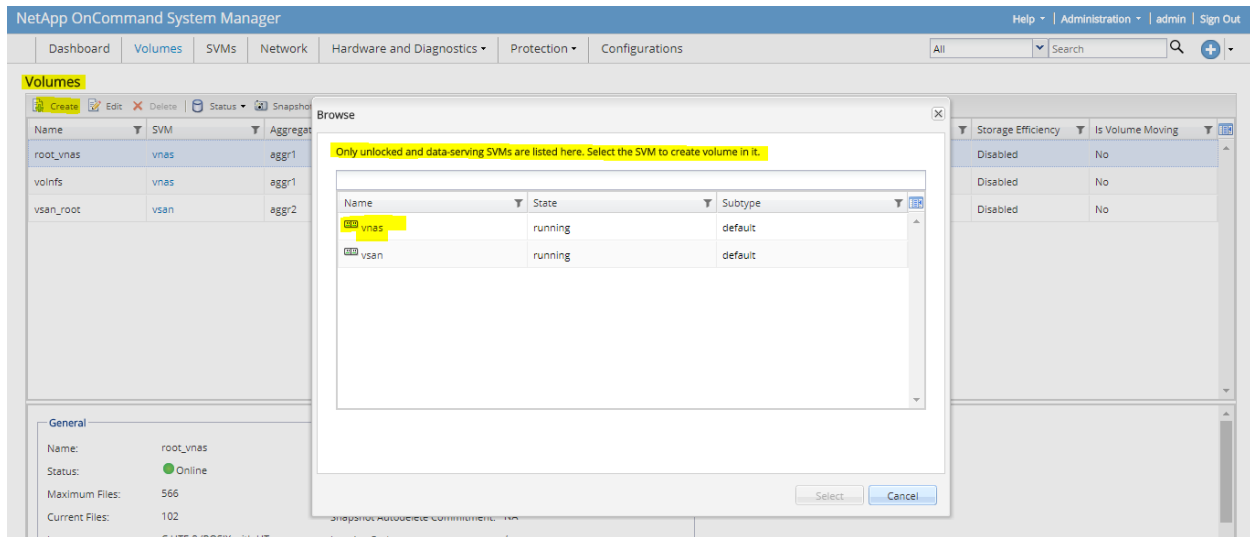
```
P90-C::> volume create -vserver vnas -volume volnfs -aggregate aggr1 -size 100m
-state online -security-style unix -unix-permissions ---rwxrwxrwx -junction-path
/volnfs -policy default -type RW -snapshot-policy default -foreground true

Warning: The export-policy "default" has no rules in it. The volume will
therefore be inaccessible.
Do you want to continue? {y|n}: y
[Job 28] Job succeeded: Successful

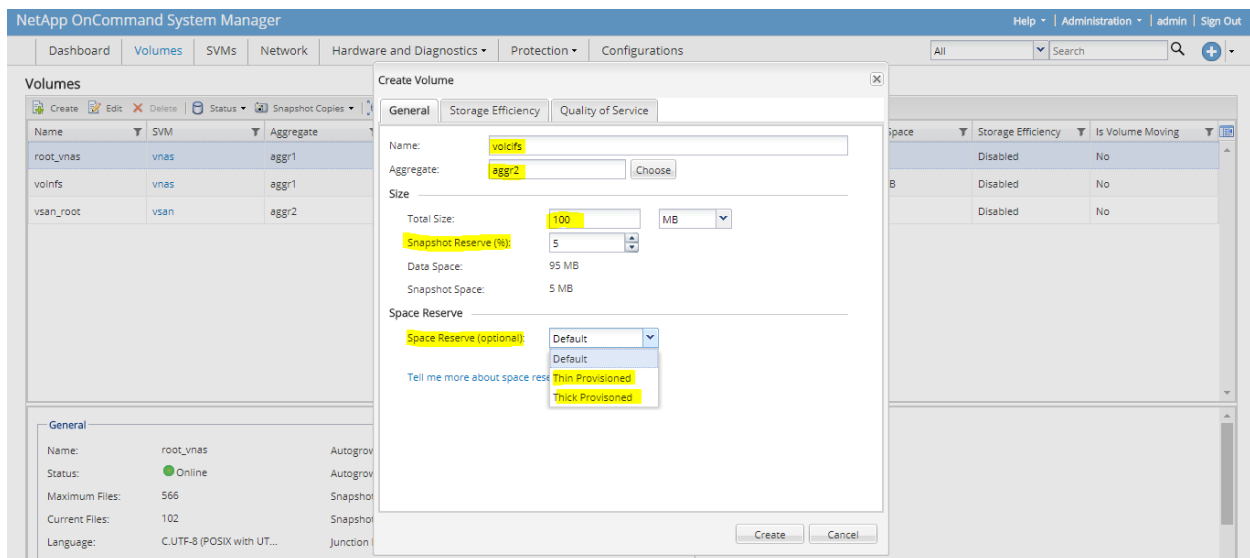
P90-C::> vol show
Vserver    Volume      Aggregate    State    Type    Size    Available    Used%
-----
P90-C-01   vol0        aggr0        online   RW      1.79GB   1.15GB      35%
vnas       root_vnas   aggr1        online   RW      20MB     18.83MB     5%
vnas       volnfs      aggr1        online   RW      100MB    94.86MB     5%
vsan       vsan_root   aggr2        online   RW      20MB     18.83MB     5%
4 entries were displayed.

P90-C::>
```

Creating a new volume using System Manager.



Volume size and properties.



The screenshot shows the NetApp OnCommand System Manager interface. The 'Volumes' tab is selected. The 'Volumes' table is displayed, showing a list of volumes with their properties. The table has columns for Name, SVM, Aggregate, Status, Thin Provisioned, % Used, Available Space, Total Space, Storage Efficiency, and Is Volume Moving. The volume 'volcifs' is highlighted in yellow.

Name	SVM	Aggregate	Status	Thin Provisioned	% Used	Available Space	Total Space	Storage Efficiency	Is Volume Moving
volcifs	vnsas	aggr2	Online	No	5	94.86 MB	100 MB	Disabled	No
root_vnas	vnsas	aggr1	Online	No	5	18.82 MB	20 MB	Disabled	No
volcifs	vnsas	aggr1	Online	No	5	94.85 MB	100 MB	Disabled	No
vsan_root	vsan	aggr2	Online	No	5	18.83 MB	20 MB	Disabled	No

Qtree Management.

Listing the qtree information.

```
P90-C::> volume qtree show
Vserver      Volume      Qtree      Style      Oplocks     Status
-----
vnas         root_vnas   ""         unix       enable      normal
vnas         volcifs     ""         unix       enable      normal
vnas         volnfs      ""         unix       enable      normal
vsan         vsan_root   ""         unix       enable      normal
4 entries were displayed.
```

Creating a new qtree.

```
P90-C::> volume qtree create -vserver vnas -volume volnfs -qtree teamq1 -security-style unix -oplock-mode enable -unix-permissions ---rwxrwxrwx -export-policy nfs1

P90-C::>
P90-C::> volume qtree show
Vserver      Volume      Qtree      Style      Oplocks     Status
-----
vnas         root_vnas   ""         unix       enable      normal
vnas         volcifs     ""         unix       enable      normal
vnas         volnfs      ""         unix       enable      normal
vnas         volnfs      teamq1     unix       enable      normal
vsan         vsan_root   ""         unix       enable      normal
5 entries were displayed.

P90-C::>
```

Creating qtree using System Manager.

NetApp OnCommand System Manager

Dashboard | Volumes | **SVMs** | Network | Hardware and Diagnostics | Protection | Configurations

vnas | Overview | Volumes | Namespace | Shares | **Qtrees** | Quotas | SVM Settings

Qtrees

Name	Volume	Security Style	Status	Oplocks	Export Policy
teamq1	volnfs	unix	normal	Enabled	nfs1

General

Name: teamq1
Status: normal
Volume: volnfs
Security Style: unix
Mount Path: /volnfs/teamq1

Export Policy Details

Export Policy: nfs1

Export Rules:

Rule Index	Client	Access Protocols	Read-Only Rule	Read/Write Rule	Superuser Access
1	10.0.0.0/8	Any	Any	Any	None

Create Qtree

Details

Quota

Name:

teamq2

Volume:

volnfs

Browse...

Oplocks:

☒ Enable oplocks for files and directories in this qtree

Security Style:

UNIX

Export Policy:

nfs1

Browse...

Create Export Policy

Tell me more about qtree options

Create

Cancel

NetApp OnCommand System Manager

Help

Administration

admin

Sign Out

Dashboard

Volumes

SVMs

Network

Hardware and Diagnostics

Protection

Configurations

All

Search

vnas

Overview

Volumes

Namespace

Shares

Qtrees

Quotas

SVM Settings

Qtrees

Create

Edit

Change Export Policy

Delete

Refresh

Name	Volume	Security Style	Status	Oplocks	Export Policy
teamq2	volnfs	unix	normal	Enabled	nfs1
teamq1	volnfs	unix	normal	Enabled	nfs1

```

P90-C::> volume qtree show
Vserver      Volume      Qtree      Style      Oplocks     Status
-----
vnas         root_vnas   " "        unix       enable      normal
vnas         volcifs     " "        unix       enable      normal
vnas         volnfs      " "        unix       enable      normal
vnas         volnfs      teamq1     unix       enable      normal
vnas         volnfs      teamq2     unix       enable      normal
vsan         vsan_root   " "        unix       enable      normal
6 entries were displayed.

```

Modifying the volume or qtree security style.

```
P90-C::> volume modify -vserver vnas -volume volcifs -security ntfs

Warning: CIFS is not configured for Vserver vnas. Files in this volume that have
        ACLs cannot be accessed or modified from NFS. Change the volume
        security style to "unix" if access from NFS is required before CIFS is
        configured.
Do you want to continue? {y|n}: y
Volume modify successful on volume volcifs of Vserver vnas.

P90-C::>
P90-C::> volume qtree show
```

Vserver	Volume	Qtree	Style	Oplocks	Status
-----	-----	-----	-----	-----	-----
vnas	root_vnas	""	unix	enable	normal
vnas	volcifs	""	ntfs	enable	normal
vnas	volnfs	""	unix	enable	normal
vnas	volnfs	teamq1	unix	enable	normal
vnas	volnfs	teamq2	unix	enable	normal
vsan	vsan_root	""	unix	enable	normal

```
6 entries were displayed.

P90-C::>
```

Listing the volume quota state.

```
P90-C::> volume quota show
```

Vserver	Volume	State	Scan Status
-----	-----	-----	-----
vnas	root_vnas	off	-
vnas	volcifs	off	-
vnas	volnfs	off	-
vsan	vsan_root	off	-

```
4 entries were displayed.
```

Listing the quota policy and rule.

```
P90-C::> volume quota policy show
```

Vserver	Policy Name	Last Modified
-----	-----	-----
vnas	default	11/16/2017 09:12:08
vsan	default	11/16/2017 09:14:55

```
2 entries were displayed.

P90-C::>
P90-C::> volume quota policy rule show
This table is currently empty.

P90-C::>
```

Creating a new quota policy.

```
P90-C::> volume quota policy create -vserver vnas -policy-name vnasqp1

P90-C::> volume quota policy show
Vserver      Policy Name      Last Modified
-----
vnas         default          11/16/2017 09:12:08
vnas         vnasqp1          11/16/2017 12:30:26
vsan         default          11/16/2017 09:14:55
3 entries were displayed.

P90-C::>
```

Creating a new quota rule.

```
P90-C::> volume quota policy rule create -vserver vnas -policy-name vnasqp1 -volume volnfs -type tree -target "teamq1" -disk-limit 25m -file-limit 1000 -threshold 22m

P90-C::> volume quota policy rule show

Vserver: vnas      Policy: vnasqp1      Volume: volnfs

Type  Target  Qtree  User Mapping  Disk Limit  Soft Disk Limit  Files Limit  Soft Files Limit  Threshold
-----
tree  teamq1  ""     -            25MB       -          1000           -          22MB

P90-C::>
```

Modifying the vservers quota policy.

```
10.3.3.50 - PuTTY

P90-C::>
P90-C::> vserver show -fields quota-policy
vserver quota-policy
-----
P90-C -
P90-C-01 -
vnas default
vsan default
4 entries were displayed.

P90-C::> vserver modify -vserver vnas -quota-policy vnasqp1

P90-C::> vserver show -fields quota-policy
vserver quota-policy
-----
P90-C -
P90-C-01 -
vnas vnasqp1
vsan default
4 entries were displayed.

P90-C::>
```

Enabling the quota for volume.

```
P90-C:~> volume quota on -vserver vnas -volume volnfs
[Job 34] Job is queued: "quota on" performed for quota policy "vnasqp1" on volume "volnfs" in Vserver "vnas".

P90-C:~> volume quota show
```

Vserver	Volume	State	Scan Status
vnas	root_vnas	off	-
vnas	volcifs	off	-
vnas	volnfs	on	-
vsan	vsan_root	off	-

4 entries were displayed.

```
P90-C:~>
```

Check from the linux server, as quota rule only we will get the space.

```
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]# mkdir /vnas_teamq1
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]# mkdir /vnas_teamq2
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]# mount -t nfs 10.2.2.31:/volnfs/teamq1 /vnas_teamq1
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/vg_linuxsenthil-lv_root	18G	5.0G	12G	31%	/
tmpfs	242M	0	242M	0%	/dev/shm
/dev/sda1	485M	30M	430M	7%	/boot
10.2.2.31:/volnfs/teamq1	25M	0	25M	0%	/vnas_teamq1

```
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]#
```

```
[root@LinuxSenthil ~]# mount -t nfs 10.2.2.31:/volnfs/teamq2 /vnas_teamq2
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]#
[root@LinuxSenthil ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/vg_linuxsenthil-lv_root	18G	5.0G	12G	31%	/
tmpfs	242M	0	242M	0%	/dev/shm
/dev/sda1	485M	30M	430M	7%	/boot
10.2.2.31:/volnfs/teamq1	25M	0	25M	0%	/vnas_teamq1
10.2.2.31:/volnfs/teamq2	35M	0	35M	0%	/vnas_teamq2

```
[root@LinuxSenthil ~]#
```

Listing the quota report information.

```
P90-C::> volume quota report
Vserver: vnas
```

Volume	Tree	Type	ID	-----Disk----- Used Limit	-----Files----- Used Limit	Quota Specifier
volnfs	teamq1	tree	1	0B 25MB	1 1000	teamq1
volnfs	teamq2	tree	2	0B 35MB	2 2000	teamq2

2 entries were displayed.

```
P90-C::>
```

NetApp OnCommand System Manager

Help | Administration | admin | Sign Out

Dashboard | Volumes | SVMs | Network | Hardware and Diagnostics | Protection | Configurations

vnas | Overview | Volumes | Namespace | Shares | Qtrees | Quotas | SVM Settings

Quotas

User Defined Quotas | Quota Report | **Quota Status on Volumes**

Activate Deactivate Resize Refresh

Volume	Status	Error
root_vnas	Off	No Error
volnfs	On	No Error
volcifs	Off	No Error

Error: No Error Warning: No Warning

